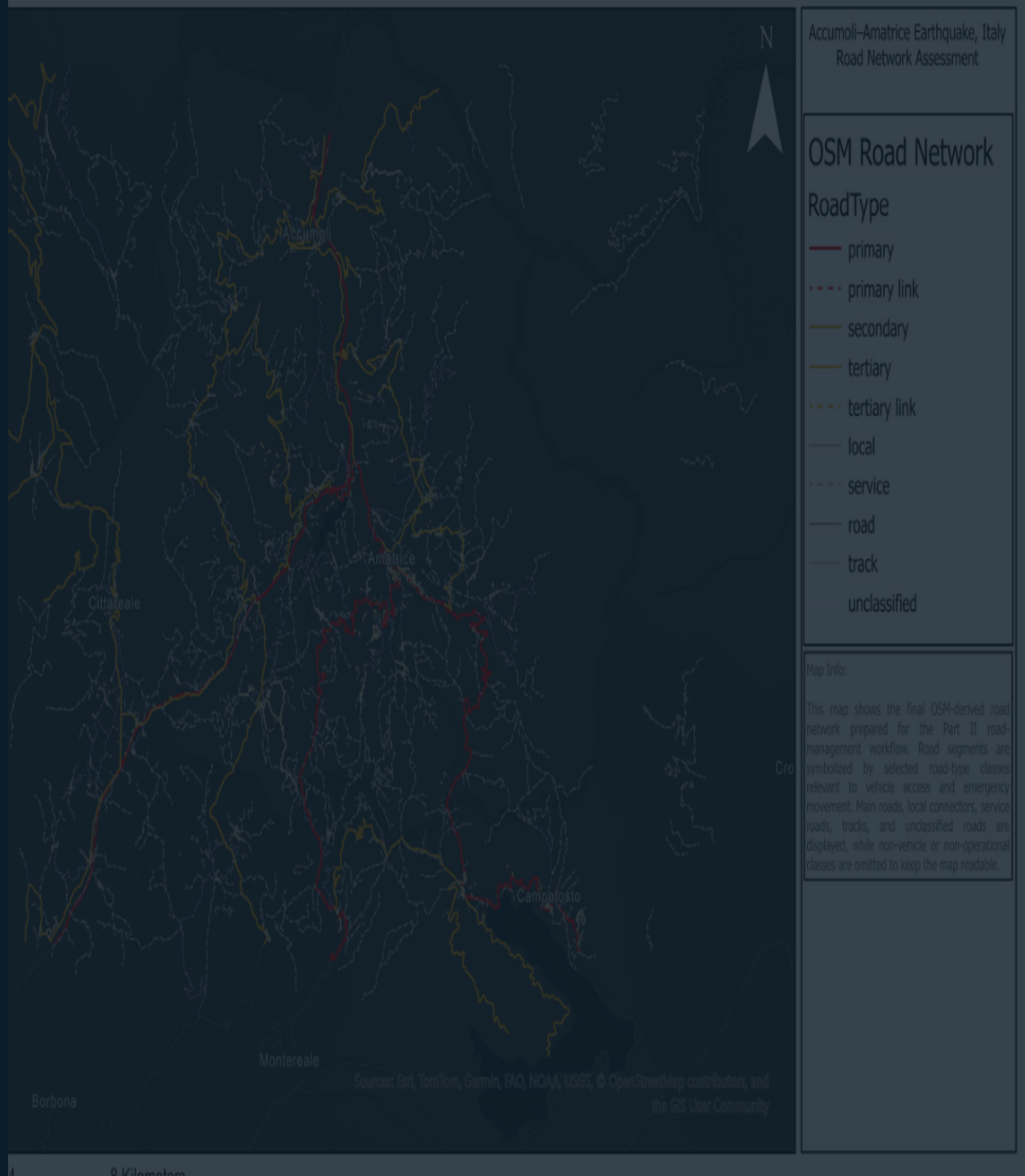
ENGINEERING PORTFOLIO

Earthquake Emergency Road Accessibility

ArcPy automation / custom Dijkstra routing / reopening priorities

2016 CENTRAL ITALY EARTHQUAKE

Politecnico di Milano | GPEM group project



From road damage to emergency access

After an earthquake, road availability depends on more than whether a road is mapped.

01

Evidence

Obstructions, damage severity and confidence vary by road section.

02

Connectivity

Local roads and tracks may preserve alternatives where major roads fail.

03

Response

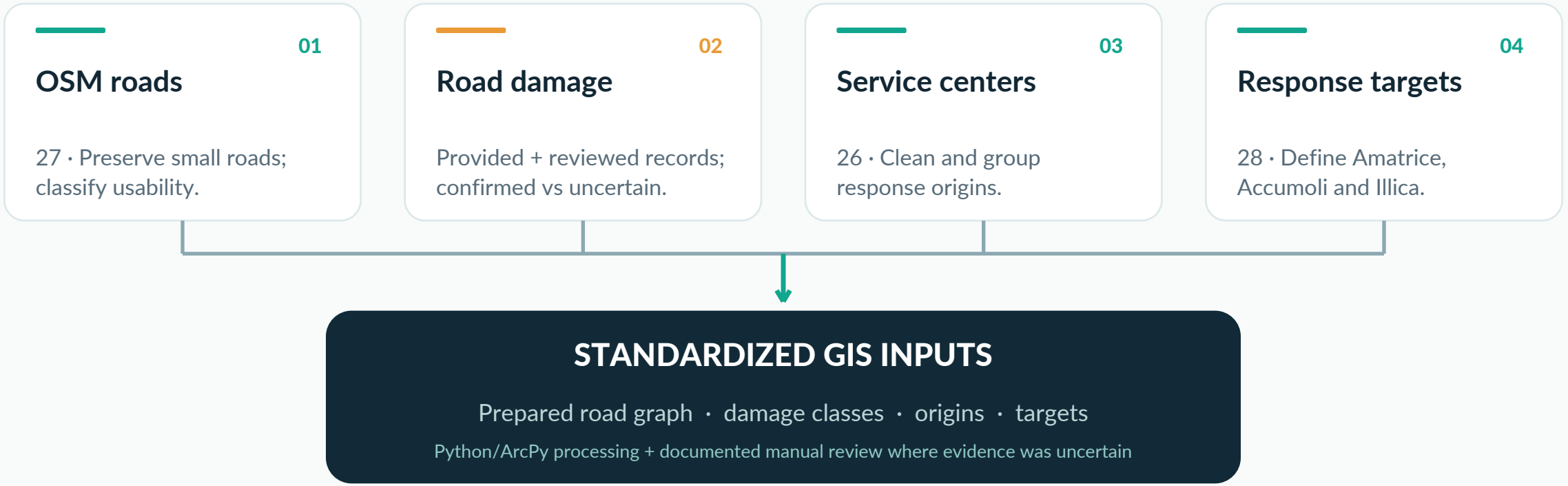
Service facilities need viable routes to Amatrice, Accumoli and Illica.

DECISION-SUPPORT GOAL

Assess selected emergency corridors and prioritize verified damaged sections for reopening.

Prepare a reliable analytical base

Four input streams are standardized before routing. The original damage evidence process is described, not republished as ML code.



Routing to decisions: one connected pipeline

Dijkstra uses a weighted graph; post-processing turns routes into corridor and reopening decision layers.

01
Graph + edge rules
Length, passability, damage confidence

02
Dijkstra routing
Conservative and degraded paths

03
Response corridors
Access condition per selected road link

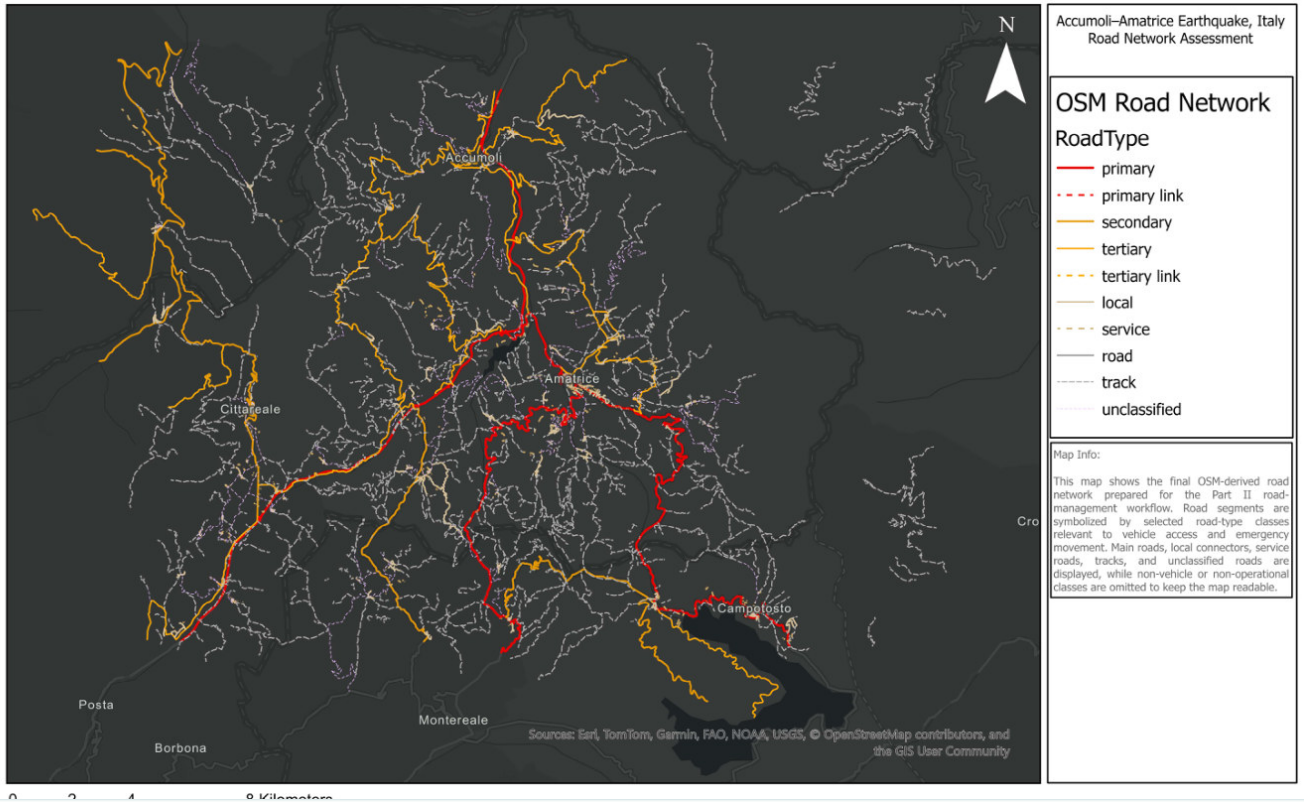
04
Reopening priority
Confirmed damage on response corridors

PUBLISHED OUTPUTS

GIS maps + target access status + classified corridors + ArcGIS FeatureServer

Keep local connections instead of over-filtering

An earlier road-cleaning strategy removed minor roads. The revision preserved full OSM extraction and classified operational



1. **Initial problem**
   A simplified network lost secondary and rural links.

2. **Correction**
   Keep geometries; assign road type, width proxies and eligibility.

3. **Why it matters**
   Alternative access options remain in the connectivity graph.

Figure source: updated academic report, p. 23

Damage-aware routing, not traffic simulation

Cost is based on road-segment length and a damage/uncertainty multiplier for eligible links.

ACTUAL EDGE COST — **length × damage / uncertainty multiplier**

No confirmed damage	×1.0
Class 1 damage	×1.5
Class 2 damage	×5.0
Unverified only	×2.0
Class 3 damage	blocked / ×100

SCENARIO CHOICE

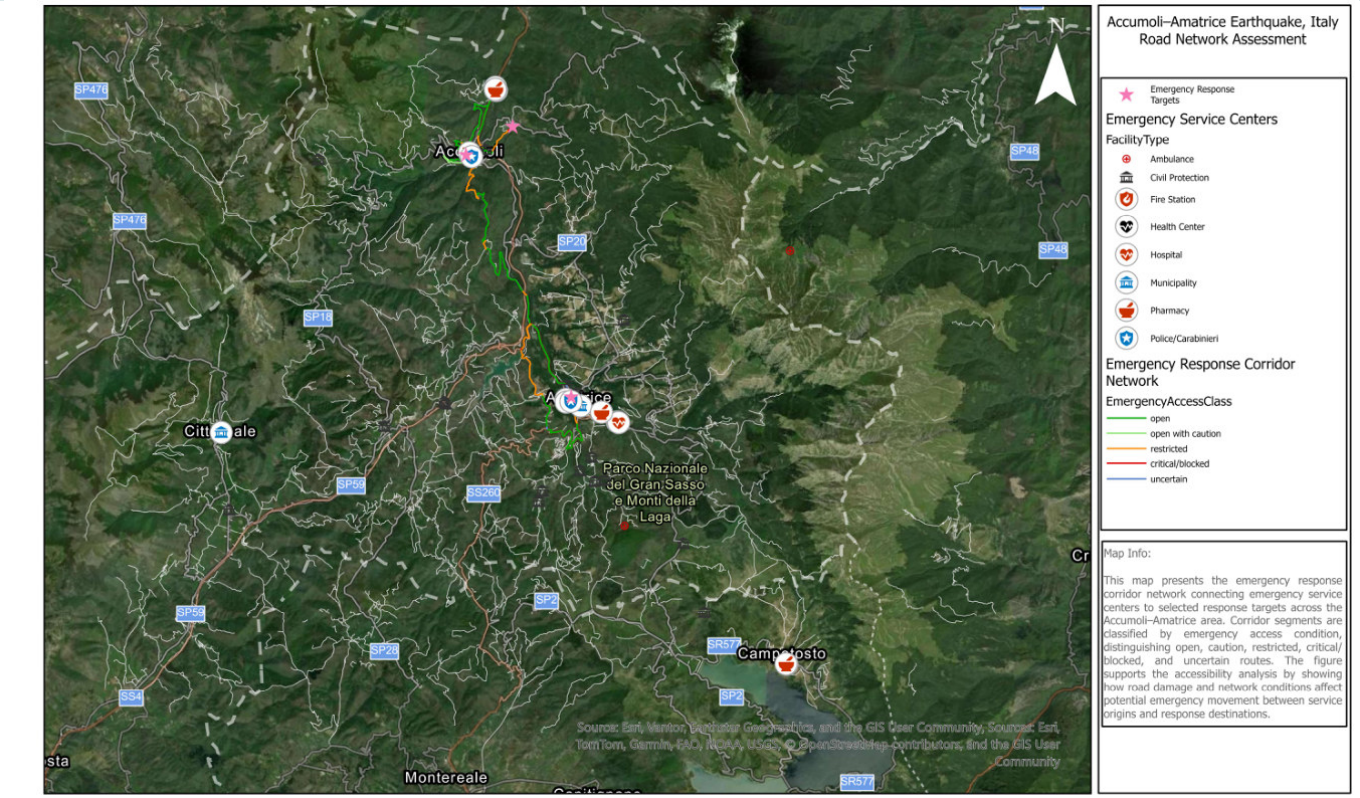
Conservative
Class 3 segments are excluded.

Degraded
Class 3 links carry an extreme penalty; this is not evidence of passability.

Road width informs subsequent access classification; it is not a separate multiplier in the published routing function.

Routes become response corridors

Selected origin-to-target paths are converted into road sections, preserving damage and access attributes.



CORRIDOR ACCESS CLASSES

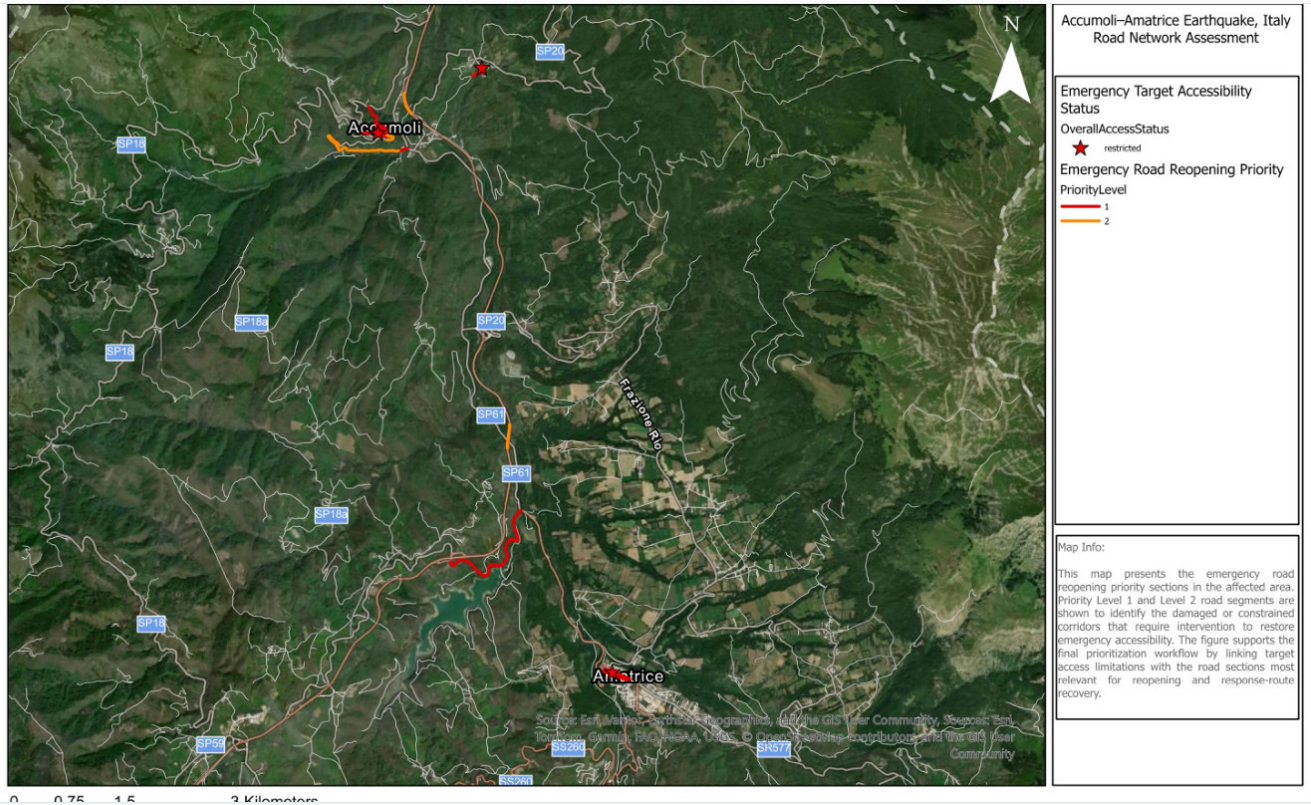
- Open
- Open with caution
- Restricted
- Critical / blocked
- Uncertain

These are scenario-based access interpretations, not confirmed live road conditions.

Figure source: updated academic report, p. 37

Reopening priorities from damaged corridors

Priority is assigned to confirmed damaged sections overlapping selected emergency response corridors.



PRIORITY 1

Critical blockage or very limited width

PRIORITY 2

Restricted passage or medium damage

PRIORITY 3

Lower immediate corridor impact

Unverified sections remain separate.

Figure source: updated report, p. 41 (visible mapped priorities 1 and 2)

A published ArcGIS FeatureServer

The spatial data are published as a service. A fully configured operational web map or dashboard is a separate stage.

01
Base context

Road network, service centers and targets

02
Damage evidence

Ground-truth, confirmed and validation-needed layers

03
Accessibility

Corridors and target access status

04
Recovery

Road reopening priority and support layers

PUBLISHED DATA SERVICE

GPEM_ROAD_MANAGEMENT / FeatureServer

Open service directory (click here in PDF)

Source: updated report, pp. 44–46

From a technical workflow to inspectable evidence

01

Connectivity

Kept small OSM roads instead of sacrificing usable graph links.

02

Transparency

Dijkstra penalties and scenario distinctions are explicit in code.

03

Traceability

Routing results become classified corridors and reopening priorities.

CODE + DOCUMENTATION

github.com/amirshokri17/seismic-emergency-road-accessibility

PUBLISHED ARCGIS FEATURE SERVICE

Open service directory

Academic group project: Amirhossein Shokri, Nima Khamsehzadeh, Sildev Kumar Ray, MohammadReza Abolhasani.